Cross-Lingual Multi-Label Emotion Classification Using Large Language Models

COURSE PROJECT BRIEF

Course	Introduction to Large Language Models (LLMs): Course Project
Groups	Three students per group
Duration	Weeks 1–8 of the course
Deliverables	Report, codebase, and presentation

1. Objective & Motivation

Large Language Models are strikingly capable at language understanding and generation, but how well that extends to multi-label tasks across many languages and to domain-specific data like emotion in dialogue is still an open research question. Emotion is also expressed differently from one language and culture to the next, which makes cross-lingual emotion detection a demanding test of what these models can really do.

This project investigates how well LLMs classify emotions across languages using the XED multilingual emotion dataset. You will compare two adaptation approaches, **prompt engineering and parameter-efficient fine-tuning (PEFT)** and examine their cross-lingual transfer to an assigned language. You will assess where the models succeed and struggle, including prompt sensitivity, computational cost, differences in pretrained language coverage, and limitations introduced by projected annotations.

“Multi-label” means that a single subtitle line can express several emotions at once—for example, both surprise and fear—rather than exactly one emotion. This makes the task more complex than single-label classification and requires the multi-label evaluation metrics defined in §7.

2. Dataset

The XED dataset (Helsinki-NLP/XED on GitHub) provides emotion-annotated movie subtitles.

- **Multilingual:** English and Finnish contain human annotations; most additional-language data were created by projecting labels to aligned subtitle lines. The projected labels should therefore be treated as projected reference labels rather than independent human judgements in each language.
- **Multi-label:** Each subtitle line can be associated with multiple emotions from Plutchik's eight core categories: anger, anticipation, disgust, fear, joy, sadness, surprise, and trust.
- **Sentence-level subtitles:** Each example is a subtitle line. Surrounding dialogue context is not part of the standard input and may only be introduced as a clearly documented optional extension.

The available projected-language files vary enormously in size. At the low end, Macedonian has 300 labelled lines, with Vietnamese (956), Slovak (975), and Icelandic (977) close behind. At the high end, Brazilian Portuguese has 12,295 lines and Spanish 11,303—more than a 40-fold difference. The language allocation in section 3 is designed to expose the cohort to this variation.

3. Methodology & Tasks

You will work with open-weight language models that can be run through tools such as Hugging Face Transformers. Select models with suitable multilingual coverage and a size that can be prompted and fine-tuned using the available course infrastructure.

Language Allocation

Each group works on English plus one assigned XED language. Across the cohort, assignments should cover a range of dataset sizes from a few hundred labelled lines to more than twelve thousand. The cohort may then explore whether the observed gap between fine-tuning and prompting is associated with the amount of target-language training data.

The languages below are grouped into five resource bands. The examples are illustrative rather than exhaustive.

Band	Approx. annotated lines	Example languages
1 — extreme low	under ~1,000	Macedonian (300), Vietnamese (956), Slovak (975), Icelandic (977)
2 — very low	~1,000–4,400	Danish, Russian, Bosnian, Slovenian, Arabic, Norwegian
3 — mid-low	~4,400–6,000	Hebrew, Swedish, Dutch, German, Hungarian, Croatian
4 — mid-high	~6,000–7,300	Czech, Italian, Bulgarian, Polish, Portuguese, French
5 — high	~8,000 and above	Greek, Finnish, Serbian, Turkish, Romanian, Spanish, Brazilian Portuguese

How languages are assigned

- Groups are distributed across the five bands so that the cohort covers the broadest feasible range.
- At the start of Week 1, a short survey collects which of these languages, if any, someone in your group can already read.
- Where a group has a reader for a language in its band, that preference is taken into account; the instructor confirms the final assignments and settles any overlaps between groups.

> **If no one in your group reads the assigned language, that is not a problem for quantitative evaluation.** The assigned-language outputs are evaluated against XED's projected reference labels. You may inspect aligned English lines to help interpret examples, but those lines do not independently validate translation quality, cultural appropriateness, or the emotional meaning of the assigned-language subtitle.

Required Data-Splitting Protocol

- Create one fixed 70/20/10 train/development/test split using a documented seed and a multi-label stratification procedure.
- Split by XED alignment identifier before constructing the English and assigned-language datasets. Parallel versions of the same subtitle must remain in the same partition.
- For the required experimental matrix, use matched English–assigned-language pairs: the English dataset should contain the English lines aligned with the assigned-language examples.
- Detect exact duplicate lines and ensure that duplicates do not occur across partitions.
- Use the development set for prompt selection, hyperparameter selection, checkpoint selection, and all other design decisions. Use the test set only for the final reported evaluation.
- Draw all few-shot demonstrations from the training set only.

Minimum Required Experimental Matrix

Use one **primary base model** for all prompting and PEFT conditions so that the comparisons remain interpretable. Use a **second model only for the zero-shot and few-shot prompting conditions**. The following experiments are mandatory:

Condition	Model(s)	Training / Demonstrations	Evaluate on
Zero-shot prompting	Primary + second	None	English + assigned language
Five-shot prompting	Primary + second	Five fixed, aligned training examples in the evaluation language	English + assigned language
English-only PEFT	Primary	English training set	English + assigned language
Target-language PEFT	Primary	Assigned-language training set	Assigned language
Bilingual PEFT	Primary	English + assigned-language training sets	English + assigned language

One PEFT method—LoRA or QLoRA—is required. A second PEFT method, additional ranks/alpha values, translated instructions, context windows, or controlled downsampling are optional extensions. Use the development set to select one final PEFT configuration; a full hyperparameter grid is not required.

Phase 1: LLM Adaptation and Fine-tuning (Weeks 1–5)

- **Data preparation:** Format the English and assigned-language training partitions as instruction-following or conversational examples. Use exactly the eight permitted emotion names and preserve the multi-label targets.
- **Parameter-efficient fine-tuning:** Use LoRA or QLoRA to run the three required PEFT conditions in the experimental matrix: English-only, assigned-language-only, and bilingual fine-tuning.
    - Document the adapter targets, rank, alpha, dropout, learning rate, batch size, number of epochs, maximum sequence length, quantization settings, checkpoint-selection rule, random seed, and hardware.

    - Use the development set—not the test set—to select the final configuration.
- **Evaluation:** Evaluate each PEFT condition on the test partitions specified in the matrix and compare it with the prompting conditions for the same primary base model. The SVM/BERT values reported by Öhman et al. (2020) may be discussed as historical reference points, but they must not be treated as directly comparable baselines unless the data, splits, label space, and metric definition are reproduced.
- **Qualitative analysis:** Analyze concrete test examples where fine-tuning improved or worsened predictions compared with prompting. Distinguish model errors from possible subtitle-alignment or label-projection limitations.

Phase 2: Zero-Shot and Few-Shot Prompt Engineering (Weeks 5–6)

- **Model selection:** Use the primary model selected for PEFT and one second model for prompting only. Justify both choices in terms of model size, multilingual coverage, licence, and accessibility.
- **Prompt design:** For each model, implement one zero-shot prompt and one five-shot prompt. Both prompts must:
    - define the eight permitted emotion labels and explain that one or several labels may be returned;
    - require a JSON list containing only the permitted English label names;
    - use the same English-language instruction template for English and assigned-language utterances;
    - use the same five fixed alignment IDs throughout five-shot evaluation, presenting the English examples for English evaluation and their assigned-language counterparts for assigned-language evaluation; and
    - report the complete prompt, decoding parameters, and demonstration-selection procedure.
- **Output parsing:** Normalize case and whitespace, but do not map arbitrary synonyms to labels. Remove duplicate permitted labels. Treat malformed outputs or outputs containing no permitted label as an empty predicted label set; report the malformed-output rate separately.
- **Evaluation:** Evaluate both prompt conditions on English and the assigned language using section 7. Compare models, languages, and prompt conditions, and analyse sensitivity to prompt wording on the development set. Do not redesign the prompt after inspecting test results.

4. Project Timeline & Syllabus Alignment

How the two project phases line up against the course's weekly lecture content.

Week	Lecture Topic	Project Activity
1	Introduction to LLMs	Phase 1 kickoff: group formation, language-band assignment (§3), dataset exploration, data prep begins.
2	Architectures & Scaling Laws	Phase 1 continues: environment setup, baseline model selection.
3	Training LLMs	Phase 1: fine-tuning experiments begin (SFT concepts now covered).

4	Efficiency in LLMs (LoRA/PEFT/QLoRA)	Phase 1 core work: implement the required English-only, target-language, and bilingual PEFT conditions.
5	Prompt Engineering	Phase 1 wraps up (evaluation) — Phase 2 begins (prompt design).
6	Knowledge and Reasoning	Phase 2: cross-lingual prompt transfer evaluation and analysis.
7	Multimodal LLMs & Future Directions	Course report, codebase finalization, presentation prep, submission.

5. Expected Outcomes / Deliverables

Each group produces three deliverables, all due at the end of Week 7:

- **Technical report (8–12 pages):** a written account of the methods, results, and analysis. See §6 for the required structure.
- **Reproducible codebase:** well-structured, documented, and runnable code for data preparation, prompting, PEFT, parsing, and evaluation, hosted on GitHub. It must include a README, environment or dependency file, configuration files, fixed seeds, and commands for reproducing the principal results. Do not commit model weights, access tokens, or unnecessary copies of the dataset.
- **Project presentation:** a concise presentation summarising the methodology, key findings, limitations, and conclusions.

6. Report Structure

The technical report should be 8–12 pages, excluding references and appendices, and cover the elements below. Treat this as a scaffold rather than a fixed form: you may merge or reorder sections where this makes your argument clearer, as long as a reader can find each element. The page figures are rough guidance for the 8–12-page budget, not hard limits.

Section	~Pages	What it covers
1. Introduction & Motivation	~1	The problem, why multilingual and multi-label emotion detection is hard, your objectives, and your group’s assigned language and band.
2. Background	~1	A short account of XED and the approaches you use (PEFT, prompting). Keep it brief — depth belongs in later sections.
3. Data & Task Setup	~1	The XED data for your language, projected-label status, label space, leakage controls, train/development/test split, and data formatting.
4. Methodology	~2	The models and selection rationale; PEFT setup and training conditions; prompt templates and demonstrations; decoding settings; and output parser.

5. Evaluation Protocol	~0.5–1	The metrics in section 7, random seeds, evaluation script, malformed-output policy, and controlled comparisons. Published XED values may be included separately as historical reference points.
6. Results	~2	Your quantitative findings in clear tables and charts, using the multi-label metrics.
7. Analysis & Discussion	~2	Error analysis with concrete examples; cross-lingual transfer, prompt sensitivity, and the PEFT-versus-prompting trade-off; malformed outputs; projected-label limitations; and other biases or limitations. Do not attribute cross-language differences to culture without direct evidence.
8. Conclusion & Future Work	~0.5	What you conclude, and what you would do with more time.
9. References (+ optional Appendix)	—	Full references (any length). An appendix may hold the prompts you used, extra tables, and similar material.

7. Evaluation Metrics

- **Primary:** micro-averaged F1 and sample-averaged Jaccard index.
- **Secondary:** macro-averaged F1, micro-averaged precision and recall, and Hamming loss. Remember that lower Hamming loss is better.
- **Diagnostic:** per-emotion precision, recall, F1, and support; exact-match ratio; and malformed-output rate for prompted models.

Compute every metric from the same binarized eight-label predictions. Use a fixed label order and report the software implementation and version. When a metric is undefined because a class has no positive examples or predictions, use a documented zero-division policy consistently. Report the random seed for every run; if conditions are repeated, report the mean and standard deviation.

> Absolute performance is not graded directly. Assessment should reward methodological correctness, reproducibility, interpretation, and critical analysis. See the companion grading rubric for how these map to report, codebase, and presentation scores.

8. Tools & Technologies (Recommended)

- Programming Language: Python
- Libraries:
  - Hugging Face Transformers: for accessing and working with LLMs.
  - Hugging Face PEFT: for Parameter-Efficient Fine-Tuning (LoRA, QLoRA).
  - scikit-learn: for evaluation metrics.
  - pandas and NumPy: for data handling.
  - Matplotlib and seaborn: for visualisation.

- Hugging Face Datasets: for data loading and processing.